



Find the Loop Bug

Ontario Math C3.2 — read, find, and fix code

Name: _____

Date: _____

★ I can find the bug in a loop and fix it.

Coders fix bugs

A bug is a mistake in code. Coders read carefully to FIND the bug, then FIX it — like finding a wrong step in a set of directions.

A loop bug is usually one of two things: the wrong repeat number, or a wrong block inside. Help Bit find and fix them.

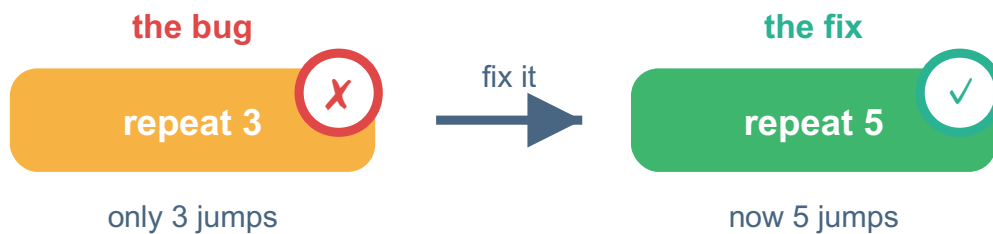
Bit should JUMP 5 times — but this loop is buggy

when green flag clicked

repeat 3

jump

Bit only jumps 3 times. The repeat number is wrong — it should be 5.



Find the wrong number, then fix it: repeat 3 → repeat 5.

1 Find the bug

Bit should jump 5 times, but the loop says repeat 3. What is wrong? What should the repeat number be? _____

2 Fix the inside

Bit should stamp red, blue. But the loop's inside is: stamp red, stamp red. Which block is wrong, and what should it be?

3 Fix the count

Bit should move 30 steps in total, 10 each time. The loop says repeat 2. What repeat number fixes it?

4 Challenge

This loop is repeat 4 with TWO "move" blocks inside. Bit moved 8 times but should move only 4. Is the bug the repeat number or the inside blocks? Fix it and explain.

Big idea

To debug a loop, first say what it **SHOULD** do, then read what it **DOES**. The bug is usually the repeat number or a block inside — change the one that is wrong.